

Shivam Panda

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EDUCATION

University of Oxford, University College

MPhys Physics (4-year), Expected July 2029

Oxford, UK

Oct 2025 – Present

- **Preliminary Examinations:** 3rd in cohort of 166 · Commendation for Practical Work · 2026 Classical Mechanics Hall of Fame (Prof. Armin Reichold)

International Academy East

IB Diploma, 42/45 · Valedictorian · Chemistry Department Award

Troy, MI

Aug 2022 – May 2025

- FIRST Robotics Competition (Technical Captain) · Dean's List Semi-Finalist

Washtenaw Community College (dual-enrolled)

4.0 GPA · Multivariable Calculus, Differential Equations, Linear Algebra

Ann Arbor, MI

Aug 2023 – Dec 2024

RESEARCH EXPERIENCE

Lattice Gauge Theory · Diffusion Models for Configuration Generation

Research with Jinchen He (Fermilab) and Yong Zhao (Argonne National Laboratory)

Summer 2026

- **Built a gauge-covariant score-based diffusion model** that performs one inverse renormalisation-group step, lifting an $L \times L$ configuration to $2L \times 2L$ at $\sim 4\times$ the coupling to precondition the HMC start in the correct topological sector.
- **Climbed a ladder from a 16×16 base to 128×128 at $\beta \approx 1660$** in 2D U(1) and U(2), matching the exact closed form to 1 ppm on the plaquette against a cold start's 51 ppm over the same budget.
- **Cut equilibration by factors of 19, 22 and 10** at $L = 32/64/128$ against a cold start. First author on the resulting paper draft; PyTorch, single-GPU workstation.

Northwestern University, CIERA

Undergraduate Researcher — advised by Prof. Wen-fai Fong

Evanston, IL

Summer 2026

- **Replaced the Monte Carlo core of the kilonova scorer with a closed form** — a Gaussian-KDE resampling estimator became an exact error-function sum — making scores deterministic and $27\text{--}32\times$ faster per call.
- **Accelerated the full scoring pipeline $6.4\times$** ($1,746\text{ s} \rightarrow 273\text{ s}$ per event) with byte-identical output across all 457 candidate scores, and swapped inverse-variance weighting for a Stouffer combiner that stays calibrated under any weighting.
- **Contributed to TROVE**, the gravitational-wave counterpart vetting platform, through to its beta release; presented the distance-scoring work at the Rubin Community Workshop.

QUANTITATIVE EXPERIENCE

Golden Road · GPU Compute Futures & Perpetuals Exchange

Quantitative Research Assistant

Remote

Jun – Aug 2026

- **Quantitative research** for an early-stage exchange building markets for GPU compute, working closely with the founding team.
- **Research to production:** took analyses from prototype to code used in the company's products.

AITHORA · Senior-Care Intelligence Platform

Quantitative Research Assistant

Remote

Aug 2026 – Present

- **Statistical modelling** for a startup building data tools for the senior-care sector.
- **Production contributions** to the company's codebase through reviewed pull requests.

ENGINEERING AND ADDITIONAL EXPERIENCE

Oxford University Racing

Wheel Assembly Chief

Oxford, UK

Oct 2025 – Present

- **Designed the 2026 suspension and steering geometry** (320 kg, 1.62 m wheelbase): traced the previous car's heavy steering to a 15° kingpin inclination and re-cut the axis to 8° KPI and 6° caster.
- **Engineered the team's first anti-roll bar assembly**, automating a Python sweep over materials, motion ratios and geometries; raised front-axle roll stiffness under 1 g by up to 54%.

TutorChase

ESAT and Oxbridge Interview Preparation Tutor

Remote

Apr 2026 – Present

- Tutor ESAT and Oxbridge science-interview applicants: design problem sets, run mock interviews, give structured feedback.

TECHNICAL SKILLS

Python (NumPy, pandas, SciPy, statsmodels, scikit-learn, PyTorch), JavaScript/TypeScript (React, React Native), MATLAB · Git, Postgres, LaTeX, Onshape